



GALACTIC COLONIZATION

& INTERGALACTIC EXODUS VIABILITY

Prepared By:
DART MEADOW

11/20/2025

REPORT: Galactic Colonization & Intergalactic Exodus Viability

Executive Summary

This report analyzes the potential expansion of human civilization from Earth to the Milky Way and eventually to a new galaxy. The analysis contrasts the "hard deadlines" of Earth's habitability against the timeline of the Milky Way-Andromeda merger.

Key Finding: The primary threat to humanity is not the galaxy collision (4.5 billion years away), but the aging of our Sun. Humanity has a "Golden Era" of ~1 billion years to colonize the Milky Way.

Section 1: The Critical Timelines

To determine how many planets we can colonize, we must first establish the time budget available before we are forced to move.

Event	Time from Now	Risk Level	Impact
Earth Peak Capacity	~2080s (60 years)	Low	Population stabilizes at ~10.4B; resource management becomes key.
Solar Luminosity	600M - 1B Years	Critical	Sun becomes 10% brighter. Oceans evaporate. (Hard Deadline)
Andromeda Merger	~4.5 Billion Years	Moderate	Galaxies merge. Orbits disrupted. Radiation risks increase.

Section 2: The Colonization Potential (Milky Way)

Using the "Colonization Wave" model to calculate expansion before the 1-billion-year deadline.

2.1 The Assumptions

- Travel Technology: Generation Ships (0.1c speed).
- Travel Time per Hop: ~40–100 years (stars 4–10 LY apart).
- Settlement Duration: 500 years per planet to build new ships.

2.2 The Expansion Rate

- Cycle Time: 100 years (Travel) + 500 years (Build) = 600 years per hop.
- Total Time Available: 1 Billion years.
- Max Cycles: $1,000,000,000 / 600 = \sim 1.6$ million hops.

2.3 The Numbers

The "human sphere" expands at roughly 1% the speed of light.

- Radius Reachable: 10 million light-years (Theoretical).

- Milky Way Size: ~100,000 light-years.
- Conclusion: We would run out of galaxy before we run out of time.
- Potential Planets: 300 Million to 40 Billion.

Note: We do not need to "ditch" the galaxy because of a lack of planets. We would likely saturate the Milky Way long before the Andromeda collision.

Section 3: The Intergalactic Exodus

If humanity decides to leave the Milky Way, we face the void between galaxies.

3.1 The Destination: Triangulum Galaxy (M33)

- Distance: ~2.73 Million Light-Years.
- Status: Stable member of Local Group.

3.2 The Journey (Fleet Logistics)

- Fleet Type: Void Arks (Artificial Planets).
- Propulsion: Fusion/Antimatter at 0.1c.
- Travel Duration: ~27 Million Years.
- Generations: ~900,000 generations.

3.3 Feasibility Check

- Biology: Significant evolutionary drift expected.
- Hull Integrity: Ships must be "living" (self-healing) materials.

Section 4: Final Report & Recommendations

Phase 1: The Milky Way Saturation (Years 2200 – Year 1B)

Aggressive expansion to seed ~10 Billion+ worlds. Humanity becomes a Type II Civilization.

Phase 2: The Galactic Winter (Year 3B – Year 4B)

Gravitational tides disrupt orbits. Begin construction of "Void Arks" and seed fleets.

Phase 3: The Great Crossing (Year 4B+)

Departure for Triangulum. ~30 Million years in transit.

Summary Comparison

Metric	Milky Way Phase	Intergalactic Phase
Primary Constraint	Industry Build Time	Fuel/Resource Preservation
Travel Time	40-100 Years	~27 Million Years
Colonizable Targets	~40 Billion Planets	1 New Galaxy (Triangulum)
Evolution	Minor Drift	Speciation Likely

ABOUT US



DART Meadow has been a cornerstone in the community, involved in both leadership and development. We are proud to lead the way towards a more efficient future with innovative ideas, game-changing technology and transparency.

We've worked on dozens of projects aimed at providing innovative engineering solutions.